Proposal Assessment 0020

Course Name: Robotics

Course Number: 01306202

Total: 30 points (Presentation => 10 points & Report =>20 points)

Due Date: October 23, 2026 (23:59 PM)

Group Name: GM ลงเซิฟ

Members of the Group:

No.	Given Name-Surname	Student ID#	Nick Name
1	Sirapat Jongchotchatchavan	67011303	Plan
2	Thitisak Suksawat	67011359	Es
3	Priyawadee Wannasin	67011278	Clean
4	Sirirat Siritaratiwat	67011305	Buck
5	Natamon Karnpanich	67011182	Ming
6	Teerphat Phuwiang	67011326	Book
7	Puthimet Limpanatewin	67011283	Put

1. Business Model Canvas (BMC)

<table>
<tr><th>Key Partners</th><th>Key Activities</th><th>Value Propositions</th><th>Customer Relationships</th><th>Customer Segments</th></tr>
<tr><td rowspan="3">- ROS/Gazebo & OpenCV Community
- Stockfish AI developers
- University Robotics & AI Labs</td><td>AI Vision: Programming webcam for real-time piece tracking.

Kinematics (IK): Simulating the robotic arm movements.

Logic: Integrating physical board state with Stockfish AI.</td><td rowspan="3">Hybrid "Digital Twin" Chess: Play with real physical pieces while an AI camera tracks every move.

Interactive Referee: If a player makes an illegal move, the AI alerts them and the simulated arm visually resets the piece on screen.

Cost Effective Robotics: A safe, zero risk simulation testbed for learning Inverse Kinematics (IK) without buying expensive physical robot arms.</td><td>AI Companion: 24/7 responsive opponent and referee.

Co creation: User feedback from project testing.</td><td rowspan="3">Solo Chess Enthusiasts: Want to play on a real board against an AI.

Robotics Students/Labs: Need a safe "Digital Twin" testbed for Robot Kinematics.

Elderly Care Centers: Looking for physical cognitive exercises (moving real pieces) guided by AI.</td></tr>
<tr><td>Key resources

Hardware: Web camera, physical chessboard, real chess pieces, laptops.</td><td>Channels

Academic: University exhibitions & project presentations.</td></tr>
<tr><td>Key Resources

Software: Webots/Gazebo (Sim), OpenCV (Vision), Stockfish.

Human: 5-7 people</td><td>Channels

Digital: GitHub (Code repo) & YouTube (Demos).</td></tr>
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Cost Structure	Revenue Streams
Hardware Setup: Cost of physical equipment (Web camera, physical chessboard, and chess pieces). **Software & Computing:** Costs for high-performance computers or cloud servers required for 3D simulation rendering and AI Vision processing. **Labor & R&D:** Development time and engineering labor (AI Vision, Inverse Kinematics, and Game Logic).	**B2C Product Sales:** Selling the "Smart Camera + AI Software" bundle to everyday chess players who want to play against an AI using a real physical board at home. **B2B Academic Licensing:** Licensing the simulation software platform to universities as an educational toolkit for teaching Robotics (IK) and Computer Vision. **Research Grants:** Funding from academic institutions or elderly care organizations to develop the system as a cognitive stimulation tool.

Project management sheet

Legend: **[O]** = Scheduled Time, **[A]** = Owner (Lead), **[B]** = Helper (Support)

Main Project Objective - To develop a 'Digital Twin' hybrid chess system that integrates real time Computer Vision for physical board tracking with a 3D simulated robotic arm. This system aims to provide an interactive AI opponent and serve as a cost-effective, zero-risk testbed for robot kinematics.

Sub Objective 1 - Camera & Piece Detection: Watch the real chessboard from above and reliably spot which piece moved and where it landed.

Sub Objective 2 - 3D Robot Arm Simulation: Build a virtual robot arm inside a 3D simulation program that smoothly picks up and places digital chess pieces.

Sub Objective 3 - Game Rules & Assistant Features: Connect the camera, the chess engine, and the 3D robot so the system detects illegal moves and plays counter-moves in real time.

Sub Objective 4 - Delivery & Presentation: Complete all testing, record a backup demonstration video, and deliver the final presentation on time.

Sub Obj	Major Tasks	Timeline (Weeks)										Assigned Team Members						
		D 1	D 2	D 3	D 4	D 5	D 6	D 7	D 8	D 9	D 10	Plan	Es	Clean	Buck	Ming	Book	Put
2	1. Set up 3D environment & digital chess pieces	O	O									**A**	B			B		
2	2. Set up the 3D robot arm inside the simulation		O	O								B				**A**		
2	3. Program Inverse Kinematics (IK) for the arm			O	O	O						**A**	B					
1	4. Set up overhead camera & feed extraction	O	O											**A**	B			
1	5. Test pre-trained AI vision model for piece detection		O	O	O									B	**A**			
1	6. Align camera to map pieces to logical squares (A1-H8)				O	O								**A**	B			

3	7. Integrate Stockfish AI engine				O	O	O						**A**			B	B	
3	8. Add chess rule checking (warn illegal moves)						O	O								B	**A**	
3	9. Code choreography (Arm resets pieces / captures)						O	O				B	**A**				B	
3	10. System Integration (Connect Camera + AI + Sim)							O	O			B		B		B		**A**
3	11. Design and develop UI (Warning screen)						O	O									B	**A**
4	12. End-to-end game tests (Ensure smooth/no lag)								O	O		B	B	B	B	B	B	**A**
4	13. Record backup video & Deliver final presentation									O	O	B	B	B	B	B	B	**A**

Category	**Equipment/ Tool**	**Purpose in the project**
Hardware (Physical)	1. Web Camera(1080p or higher)	Mounted above the board to capture real time video feed for piece detection.
	2. Physical Chessboard & Pieces	Used as the tangible, real world interface for the human player to interact with.
	3. Overhead Camera Mount / Tripod	To stably hold the web camera directly over the chessboard to prevent perspective distortion.
	4. High-Performance Laptops	Used by developers to run the 3-D simulation and engine, AI vision rendering, and backed logic simultaneously.

Software	5. Python (Programming Language)	The primary backend language used to connect the computer vision, game logic and simulation.
	6. OpenCV (Vision Library)	Used for image processing, tracking the physical chess pieces, and mapping pixels to logical board coordinates (A1-H8).
	7. Webots / Gazebo (Simulation Engine)	Used to create the 3D virtual environment and simulate the Robotics Arm's movements (Inverse kinematics).
	8. Stockfish (Chess AI Engine)	The brain of the system, used to validate illegal moves and calculate the computer's counter-moves.
Preject Management	9. Git & GitHub	For a team code collaboration and versions control across the 7 team members.